Project

Site: Eduvos LMS
Course: Network Security Assessments
Book: Project
Printed by: Noor Dyer
Date: Wednesday, 8 April 2026, 5:59 PM

Table of contents

1. Project / Research Essay
2. Instructions to Students
3. Section A
    3.1. Scenario
    3.2. Question 1
    3.3. Question 2
    3.4. Question 3
    3.5. Question 4
4. Appendix 1: Scenario
5. Appendix 2: Question 1.1

Project

Faculty:	Information Technology
Module Code:	ITNSA2-44
Module Name:	Network Security
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Internal Moderation:	Community of Practice
Copy Editor:	Kyle Keens
Total Marks:	100

Submission Week:

This module is presented on NQF level 6.

5% will be deducted from the student's assignment mark for each calendar day the assignment is submitted late, up to a maximum of three calendar days. The penalty will be based on the official campus submission date.

Assignments submitted later than three calendar days after the deadline or not submitted will get 0%. [1]

This is a **group** project.

Each group should consist of **2 - 3 students, as decided by the lecturer.**

This project contributes 50% towards the final mark.

[1] Under no circumstances will assignments be accepted for marking after the assignments of other students have been marked and returned to the students.

Instructions to Students

1. Please ensure that your answer file (where applicable) is named as follows before submission: **Module Code – Assessment Type – Campus Name – Student Number.**
2. Remember to keep a copy of all submitted projects.
3. All work must be typed.
4. Please note that you will be evaluated on your writing skills in all your projects.
5. All work must be submitted through Turnitin. The full originality report will be automatically generated and available for the lecturer to assess. Negative marking will be applied if you are found guilty of plagiarism, poor writing skills, or if you have applied incorrect or insufficient referencing. (See the "instructions to students" book activity before this activity where the application of negative marking is explained.)
6. You are not allowed to offer your work for sale or to purchase the work of other students. This includes the use of professional assignment writers and websites, such as Essay Box. You are also not allowed to make use of artificial intelligence tools, such as ChatGPT, to create content and submit it as your own work. If this should happen, Eduvos reserves the right not to accept future submissions from you.
7. One group member should be nominated to submit the assessment on behalf of the group. Multiple submissions by various group members will result in an inflated similarity index on Turnitin.

Section A

Learning Objective

Demonstrate an understanding of network security fundamentals and how they are applicable in real-life scenarios by creating virtual environments that would mimic what real world organisations would have.

Project Topic

Network Security, network design, network configuration.

Scope

Week 1 - 7

Technical Aspects

Creating simulated virtual networks

Marking Criteria

Use discretion when marking

Scenario

Study the scenario and complete the question(s) that follow:

Please use to the updated scenario under appendix 1.

BrightSoft Solutions, a growing software services company headquartered in Durban, South Africa, operates across three dedicated floors to support its expanding teams: Developers, Business Analysts, Test Engineers, and UI Designers. With the recent addition of a staging environment for client-facing demonstrations, the organisation's network demands have become more complex, requiring strict segmentation, secure external access, and resilient routing.

1. Floors and Teams

· Floor 1: Developers (3 end-user devices) and Business Analysts (3 end-user devices), each team running its own DHCP server.

· Floor 2: Test Engineers (2 end-user devices) and UI Designers (2 end-user devices). Although they share the floor physically, logical separation via VLANs is required for security.

· Floor 3: Dedicated Staging Environment, hosting a demonstration web server accessible by external partners but restricted from most internal systems, except where limited access is needed from the development team.

2. DHCP and DNS

· Developers and Business Analysts continue managing their own DHCP servers.

· A central DNS server must provide both internal name resolution (for local devices across all floors) and external lookups (for staging demos and internet access).

3. Routers and Routing

· Business Analysts' Router: Connects Floor 1 to the DNS server, web servers, and the internet.

· Developers' Router: Connects Floor 1 to the DNS server, staging environment, and the internet.

· Shared Router: Connects Floor 2 (Test Engineers, UI Designers) and Floor 3 (Staging Environment) as a cost-saving measure.

· Routing Protocols:

o RIP remains in use on one router to support certain legacy applications.

o OSPF is introduced on the Shared Router for scalable and efficient communication with the staging environment.

4. VLANs

· UI Designers and Test Engineers must be placed in separate VLANs despite sharing Floor 2, requiring 802.1Q trunking and inter-VLAN routing via a router or Layer 3 switch.

· The Staging VLAN on Floor 3 must allow limited inbound connections from the internet for demos, while filtering traffic to prevent unauthorised access to core systems.

5. Servers

· Production Web Server: Hosts live company applications; internally accessible by all departments.

· Staging Web Server: Located on Floor 3, used for demonstrations to external partners and updates from the Development team.

· DNS Server: Handles both internal resolution for all teams and external DNS lookups.

6. Security & Access Control

· Controlled External Access: Staging web server must only accept inbound traffic from approved external IP addresses on restricted ports (HTTP/HTTPS).

· Access Control Lists (ACLs): Extended ACLs required to restrict staging access from internal VLANs and block staging-to-production communication.

· NAT/Port Forwarding: Ensure external users can reach the staging server via a public IP without exposing other services.

· VLAN Segregation: Test Engineers and UI Designers cannot access each other's VLAN except for approved shared resources (e.g., a common printer or file server).

- Monitoring: Implement logging on all routers (RIP and OSPF events included) and basic intrusion detection for unusual traffic in the staging VLAN.

7. Additional Challenges

- Subnet Planning: Allocate IP ranges for each VLAN to accommodate both current and future devices.
- Redundancy: Introduce a backup link between the Shared Router and Business Analysts' Router for failover, requiring proper RIP and OSPF route advertisement.
- DHCP Resilience: Ensure DHCP servers can handle conflicts and provide fallback in case of failure.
- Performance Optimisation: Apply trunking and inter-VLAN routing enhancements (e.g., EtherChannel on core switches) to prevent bottlenecks as traffic grows.

End of Scenario

Question 1 20 Marks

Please refer to the updated Question 1.1 in Appendix 2

1.1 Using Cisco Packet Tracer, design a complete network layout that reflects the three floors described in the scenario. On Floor 1, include three devices for the Developers and three devices for the Business Analysts. Floor 2 should consist of two devices for the Test Engineers and two devices for the UI Designers, while Floor 3 is dedicated to a single staging web server. The design must also incorporate the three routers outlined in the scenario, namely the Business Analysts' Router and the Developers' Router on Floor 1, as well as the Shared Router, which provides connectivity for Floor 2 and Floor 3. (5 marks)

1.2 Configure independent DHCP servers for the Developers and the Business Analysts on Floor 1. Each DHCP server should be capable of assigning addresses exclusively to its respective VLAN or subnet, ensuring proper isolation of traffic. Once configured, verify that all devices on Floor 1 automatically obtain valid network parameters including IP addresses, subnet masks, default gateways, and DNS details. (10 marks)

1.3 Install and configure a DNS server that is accessible from all three floors of the company network. This server must be capable of resolving hostnames for internal devices across the organisation while also supporting external lookups, which will allow access to internet resources as well as the staging environment used for client demonstrations. (5 marks)

End of Question 1

Question 2 20 Marks

2.1 On Floor 2 of BrightSoft Solutions, create two separate VLANs to ensure logical segmentation between the Test Engineers and the UI Designers, who share the same physical space. Each VLAN should be configured with its own subnet and assigned to the appropriate switch ports, ensuring that devices from each department can only communicate within their respective VLAN by default. (10 Marks)

2.2 Configure the Shared Router, or alternatively a Layer 3 switch, to provide inter-VLAN routing. This configuration should allow controlled communication between the VLANs when necessary and ensure that traffic from the Test Engineers and UI Designers can also reach common resources such as the DNS server and the staging environment. (5 Marks)

2.3 Apply an extended ACL or a comparable mechanism on the Shared Router to restrict direct communication between the Test Engineers' VLAN and the UI Designers' VLAN. The ACL should block traffic unless specific exceptions are authorised, such as access to a shared printer or file server. This ensures strict isolation while still supporting minimal collaboration where required. (5 Marks)

[Sub Total 20 Marks]

End of Question 2

Question 3 30 Marks

Study the scenario and complete the question(s) that follow:

Refer to the scenario in question 1.

3.1 In Cisco Packet Tracer, configure the Floor 3 staging environment so that approved external partners can access the staging web service over HTTP/HTTPS while maintaining strict isolation from the rest of the internal network. Your configuration should demonstrate the use of NAT or port forwarding on the edge device, together with extended ACLs that restrict inbound and lateral movement, and should permit the Development team to update the staging server without exposing additional internal resources.

Provide evidence of your configuration and policy effectiveness using screenshots or command outputs (e.g., NAT translations, ACL hit counts, successful/blocked connection tests). Your submission should clearly show how you balance external demo accessibility with internal security. (20 Marks)

3.2 Beyond the Packet Tracer build, discuss the theoretical design for integrating a partially externally accessible staging environment into BrightSoft's network. In your answer, justify the placement of the staging server (e.g., dedicated VLAN/DMZ segment), outline the required control plane and data plane policies (firewall rules, ACLs, routing considerations), and describe supporting services (DNS resolution approach, certificate management, logging/monitoring, and basic intrusion detection around the staging segment).

Explain how your design preserves least privilege for internal users, limits the blast radius of a compromise, and supports operational needs such as Development updates and partner access. (10 Marks)

[Sub Total 30 Marks]

End of Question 3

Question 4 30 Marks

Study the scenario and complete the question(s) that follow:

Refer to the scenario in question 1.

4.1 In Cisco Packet Tracer, enhance network resilience and monitoring by enabling RIP on one router for legacy compatibility and OSPF on the Shared Router for efficient routing. Show how you would introduce a failover or backup link to maintain connectivity if a primary link fails.

(15 Marks)

4.2 Explain the potential challenges BrightSoft Solutions may encounter when running both RIP and OSPF within the same network infrastructure. In your answer, consider issues such as protocol compatibility, administrative complexity, and route redistribution between the two protocols. Additionally, evaluate how these challenges could affect the introduction of a failover or backup link designed to improve network resilience. (15 Marks)

[Sub Total 30 Marks]

End of Question 4

Appendix 1: Scenario

Study the scenario and complete the question(s) that follow:

Updated Scenario

BrightSoft Solutions is a software services company headquartered in Durban, South Africa, operating across three dedicated floors. The organization has experienced rapid growth, requiring a well-planned, segmented, and secure network infrastructure to support its expanding teams and new staging environment for client demonstrations.

The company network is allocated the IP address space **192.168.0.0/24** and structured across three physical floors with interconnected routers, switches, servers, and end-user devices.

Floor 1: Developers and Business Analysts

Teams and Devices:

- Developers Team: 3 end-user devices
- DHCP server managed locally by Developers team
- Business Analysts Team: 3 end-user devices
- DHCP server managed locally by Business Analysts team

Network Connectivity:

- Each team has its own router connecting the team's devices to centralised services
- Developers' router connects to: Central DNS Server, Staging Environment (Floor 3), and the Internet
- Business Analysts' router connects to: Central DNS Server, Production Web Server, and the Internet
- Routing Protocol: RIP v2 (required for legacy applications)

Floor 2: Test Engineers and UI Designers

Teams and Devices:

- Test Engineers: 2 end-user devices
- UI Designers: 2 end-user devices
- Both teams share the floor but require logical separation via VLANs for security

Shared Resources:

- Common printer and shared file server
- Both teams can access these resources through controlled inter-VLAN routing

Network Connectivity:

- Layer 3 switch handles inter-VLAN routing with 802.1Q trunking
- Single shared router connects Floor 2 to the staging environment (Floor 3) and the Internet
- Routing Protocol: OSPF (Open Shortest Path First for scalable routing)

Floor 3: Staging Environment

Infrastructure:

- Staging Web Server:
- Used for client demonstrations and receives updates from the Developers team
- Accessible from the Internet (external partners) on HTTP/HTTPS
- Accessible from Developers team for updates and maintenance
- Restricted from Business Analysts, Test Engineers, and UI Designers

Centralised Infrastructure

Central DNS Server: 192.168.0.13

- Provides internal name resolution for all devices across all floors
- Handles external DNS lookups for internet access and staging demonstrations
- Accessible from all routers and all VLANs

Production Web Server: 192.168.0.14

- Hosts live company applications accessible by all internal departments
- Connected through the Business Analysts' router

Syslog Server: 192.168.0.15

- Centralized logging for all routers and network events

Router Interconnection and Redundancy

The three routers are interconnected to allow inter-floor communication:

- Developers' router and Business Analysts' router both connect to the Central DNS Server
- Shared router connects both Floor 2 and Floor 3 to the centralised services
- A backup link exists between the Shared router and Business Analysts' router for failover and redundancy
- All routers connect to the Internet through separate ISP connections

End of Scenario

Appendix 2: Question 1.1

Updated Question 1.1

1.1 Using Cisco Packet Tracer, design a complete network layout that reflects the three floors described in the scenario. On Floor 1, include three devices for the Developers and three devices for the Business Analysts. Floor 2 should consist of two devices for the Test Engineers and two devices for the UI Designers, while Floor 3 is dedicated to a single staging web server. The design must also incorporate the three routers outlined in the scenario. (5 marks)